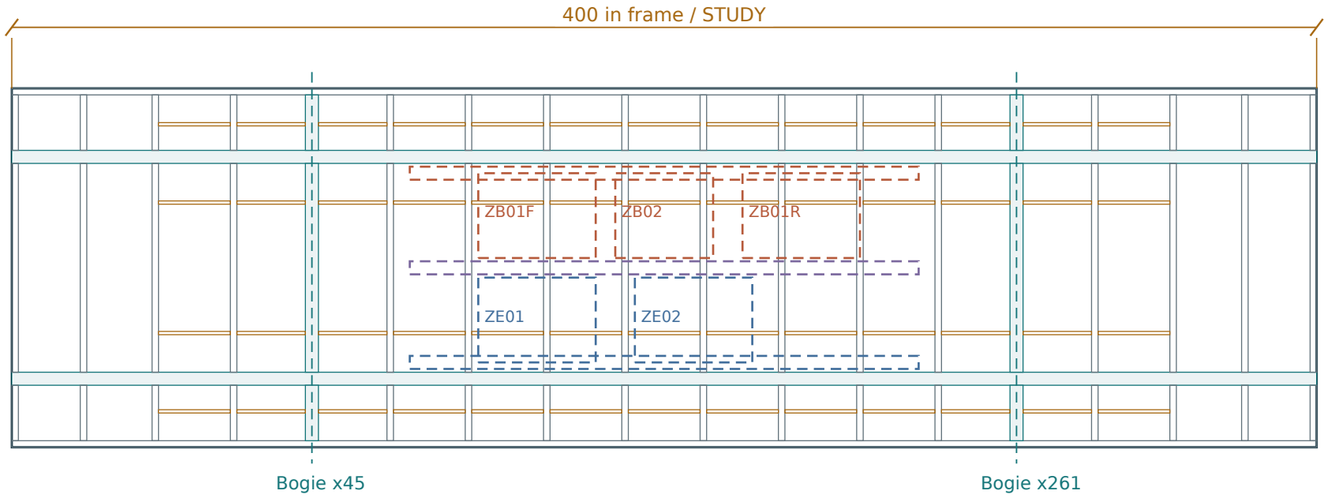
First-coach integration: installation, service access and clearance

F1 = E1 at 32 in floor, study only. Full-motion bogie clearance remains unresolved. Equipment boxes are allocated space claims, not verified products or approved mounting details.

G1 / UNDERFLOOR PLAN - PROJECTED STEEL AND LOWER EQUIPMENT



G2 / ALLOCATED BOXES - ALL PRODUCT FITS HELD

ID	Function	z [in] above rail
ZB01F /R	Two local hydraulic modules	18-25 in
ZB02	Brake reserve / DC panel	18-25 in
ZE01	AC / auxiliary conversion	18-25 in
ZE02	Separate light reserve	18-25 in
ZR01	Generic matched rail HVAC	121.61-145.61
ZR02L /R	External supply ducts	121.61-136.61

32 in STUDY - FULL-MOTION CLEARANCE UNRESOLVED

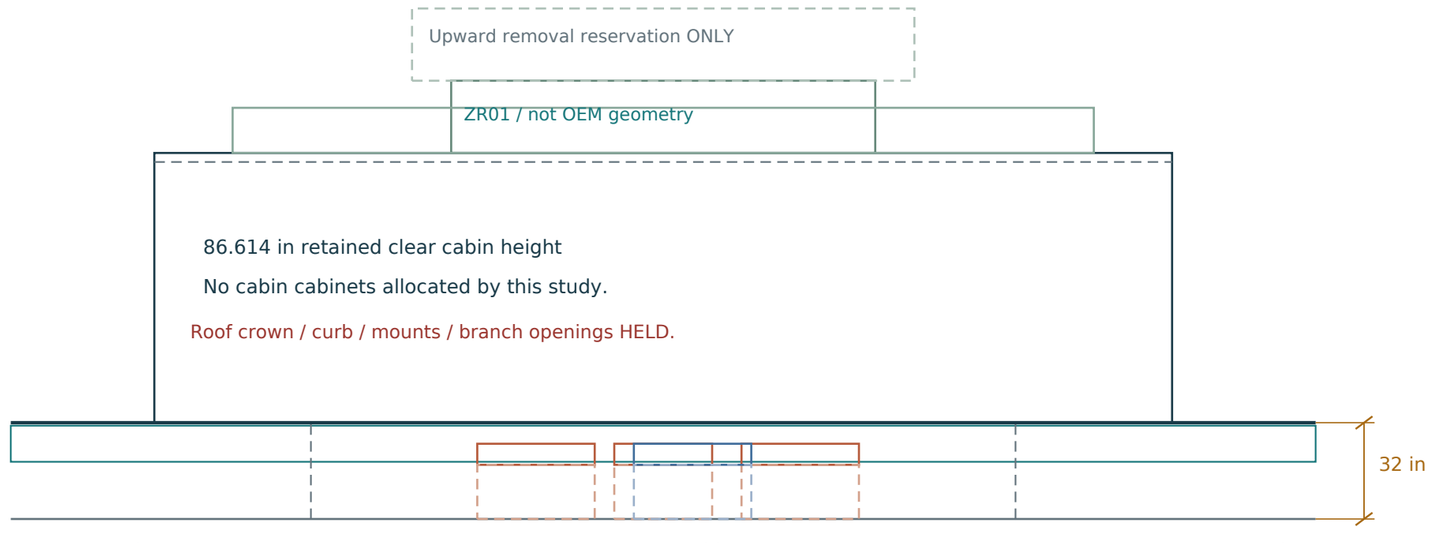
Prior E1 P3 bounding case = -24.82 mm. Negative bound denotes possible overlap, not a confirmed collision; it prevents a fit approval. Actual wheel profile and movement data are still required.

Underfloor equipment has only 7 in nominal height. No actual HCU, battery, transformer or enclosure fit is demonstrated. Five downward removal reservations extend to z0; they do not approve ground clearance or a maintenance pit.

Lower the entire box below the girder underside with approved clearance before lateral withdrawal. Service supports, vehicle restraint, lifting and isolation remain HELD. Central routing corridors terminate before bogie / bolster crossovers.

All 110 members remain no-drill / no-mount until released.

G3 / ELEVATION - OPERATING ALLOCATIONS AND MAINTENANCE SPACE



VERIFIED / source data | CALCULATED / stated inputs | STUDY / proposed geometry | HELD / required before release

PRELIMINARY ENGINEERING DEFINITION - NOT FOR FABRICATION OR CONSTRUCTION

PF-TRAIN-S3F-R0 | 2026-10-18 | References: systems_f / boarding_f / structure_f / mechanical_f

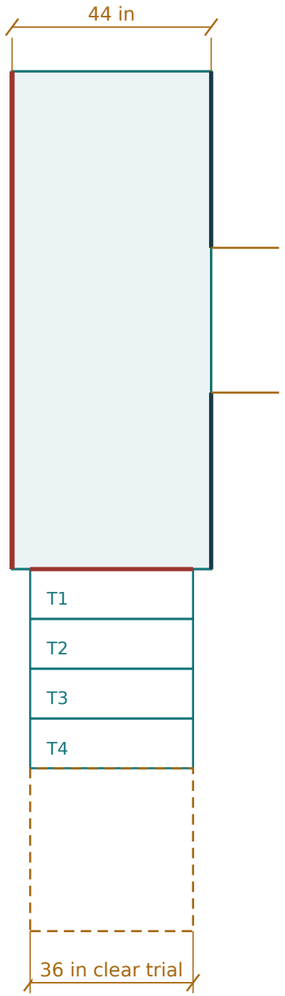
G01

17 x 11 in / R0

Regular end platform: deployable stair and station interface

F1 floor 32 in is an integration study. The 7 in rise / 11 in going benchmarks are comparisons, not an adopted vehicle code or approved staircase. Geometry is walking surfaces only.

B1 / FRONT RIGHT OPTION - PLAN / OPPOSITE SIDES ARE ALTERNATIVES



End barrier protects coupling gap.

Barrier height / attachment HELD.

End door 32 in clear / STUDY

Door / gate swings HELD.

Gate opening at y55

Positive deployed lock and

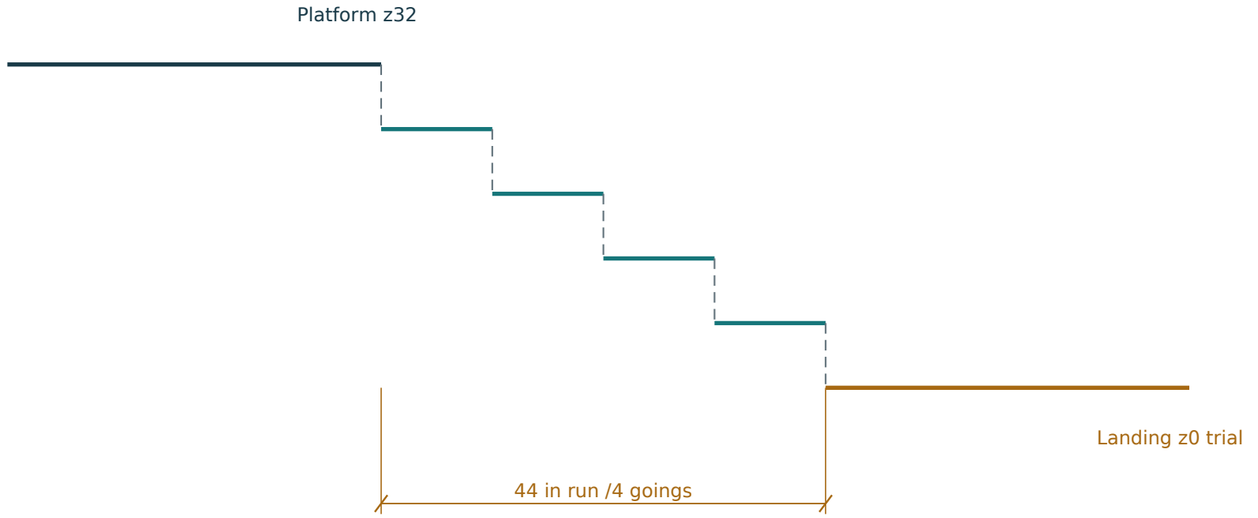
stowed retention required;

mechanism not selected.

36 x 36 lower landing trial

Actual surface elevation governs.

B2 / TRANSVERSE PROFILE - WALKING PLANES ONLY



5 equal rises x 6.4 in; tread thickness and supports HELD.

Landing z	Risers	Rise	Run
+0 in	5	6.4 in	44 in
-4 in	6	6.0 in	55 in
+4 in	4	7.0 in	33 in

A fixed 32-to-0 staircase produces a 10.4 in bottom rise if the landing drops 4 in. Match each boarding surface; do not treat rail elevation as finished terrain. No motion with stairs deployed. Handrails, guards, slip resistance and lighting remain engineered details.

R30 m adjacent bare deployed-stair separation = 2.639 in; guards, hardware, people and dynamics excluded. This does not accept stationary boarding. End-facing stairs in the30 in gap are rejected; stowed geometry HELD.

VERIFIED / source data

CALCULATED / stated inputs

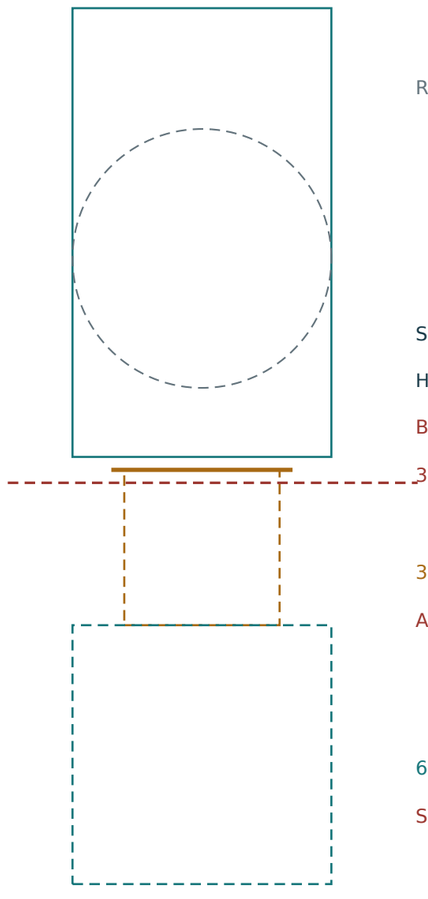
STUDY / proposed geometry

HELD / required before release

Accessible boarding: normal station, optional equipment and wayside rescue

Reservations are dimensioned for coordination; equipment selection, admitted loads, station elevations, retention and evacuation procedures remain subject to qualified review.

A1 / NORMAL STATION - RESERVATION PLAN, NOT A BRIDGE PLATE



Retained turn: R30 at x30,y6

Side door 42 in clear / STUDY

Height 78 in retained / 80 in alternative

Body face y55; station edge y58

3 in nominal gap is a sensitivity.

36 x 36 operating reservation

Actual plate size / bearing HELD.

60 x 60 station-turn reservation

Survey / terrain / station design HELD.

A2 / VERTICAL INTERFACE - MISMATCH AND BEARING

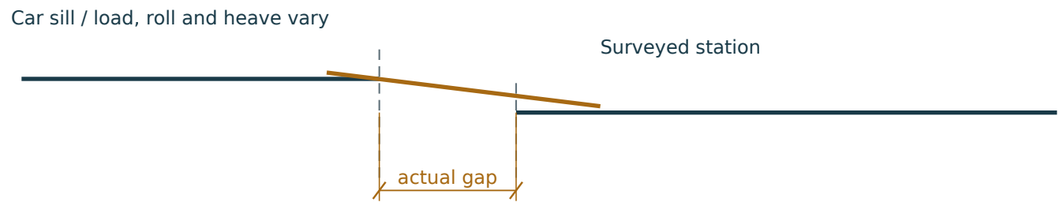
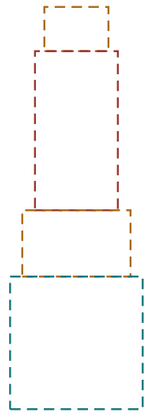


Plate bearing, thickness, stiffness, edge protection and positive retention HELD.

Use actual usable horizontal run and measured elevation mismatch. A 36 in operating reservation does not establish a 36 in bridge span or its load rating. Account for suspension travel, track/station tolerance and loaded bridge deflection.

A3 / WAYSIDE RESCUE - UNSELECTED LIFT RESERVATION



Mobilift TX dimensional comparison ONLY

The 29 in published bridge width and upper load case prevent selection. Dimensions alone do not establish accessible service. An onboard lift also needs a wider door and conflicts with the retained turn reservation.

Release a complete stalled-train method: stable receiving area, access from both travel directions, trained assistants, suitable equipment and independent recovery after loss of train power. Actual rescue apparatus and setup remain HELD.

Optional onboard lift: 45.85 in minimum door-width reference exceeds retained 42 in door; actual swept CAD and rail application acceptance are required.

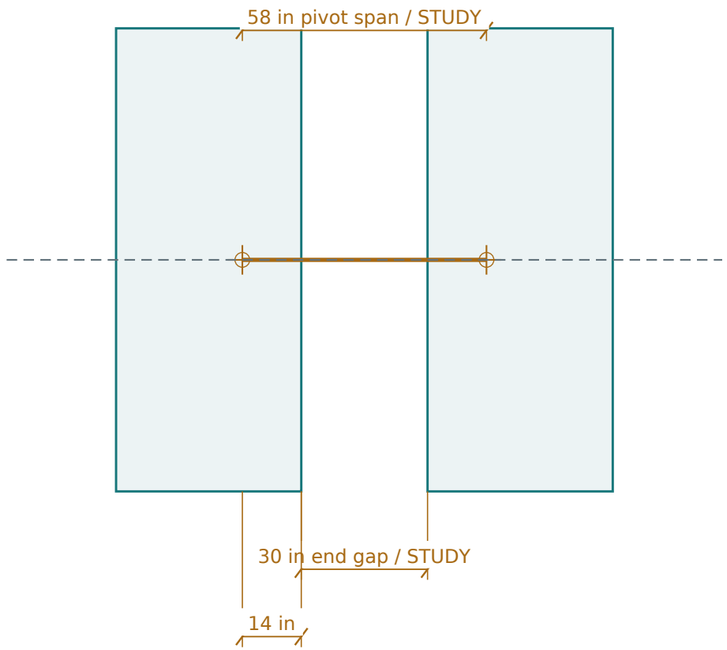
Do not qualify evacuation by carrying an occupied wheelchair down stairs. Test actual devices, attendants and admitted combined loads in a released setup.

VERIFIED / source data CALCULATED / stated inputs STUDY / proposed geometry HELD / required before release

Intercar connection: geometry, load path and retained interfaces

Plan distances are retained sensitivity values only. Actual coupler height, articulation, damping, pins, steel, welds, bolt patterns and separation retention require documented component design.

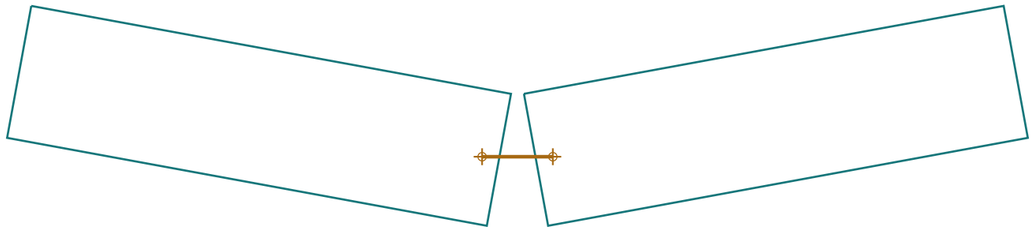
C1 / STRAIGHT PLAN - PIVOT SYMBOLS ARE NOT PIN SIZES



Platforms are protected passenger zones; no open passage over the connection.

Tension / compression enters engineered frame attachments, not deck skin.

C2 / TWO-COACH CURVE STUDY - WHOLE FRAME OUTLINES



R30 m geometry shown; constant-radius rigid-body sensitivity only.

Radius	Body angle difference	Frame separation
30 m	20.77 deg	10.63 in
50 m	12.50 deg	18.20 in
75 m	8.34 deg	22.08 in

Positive frame separation does not clear actual coupler heads, guards, hoses or passengers. Repeat with full OEM geometry, reverse curves, slack, yaw / pitch / roll and wear.

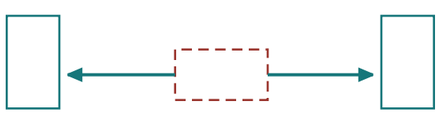
C3 / SIDE AND TRANSVERSE SECTIONS - SYMBOLIC INTERFACES

Side / not to scale



Hc, clearances and retention = OEM

Section / load distribution held



Engineered anchor / diaphragm to girders

C4 / FUNCTIONAL EXPLODED VIEW - NOT PART GEOMETRY



Separate functions: service-jumper support and passenger exclusion guard.

Functional order only; return actual exploded CAD and retention details.

VERIFIED / source data CALCULATED / stated inputs STUDY / proposed geometry HELD / required before release

PRELIMINARY ENGINEERING DEFINITION - NOT FOR FABRICATION OR CONSTRUCTION

PF-TRAIN-S3F-R0 | 2026-10-18 | References: mechanical_f / symbolic interface topology

C01

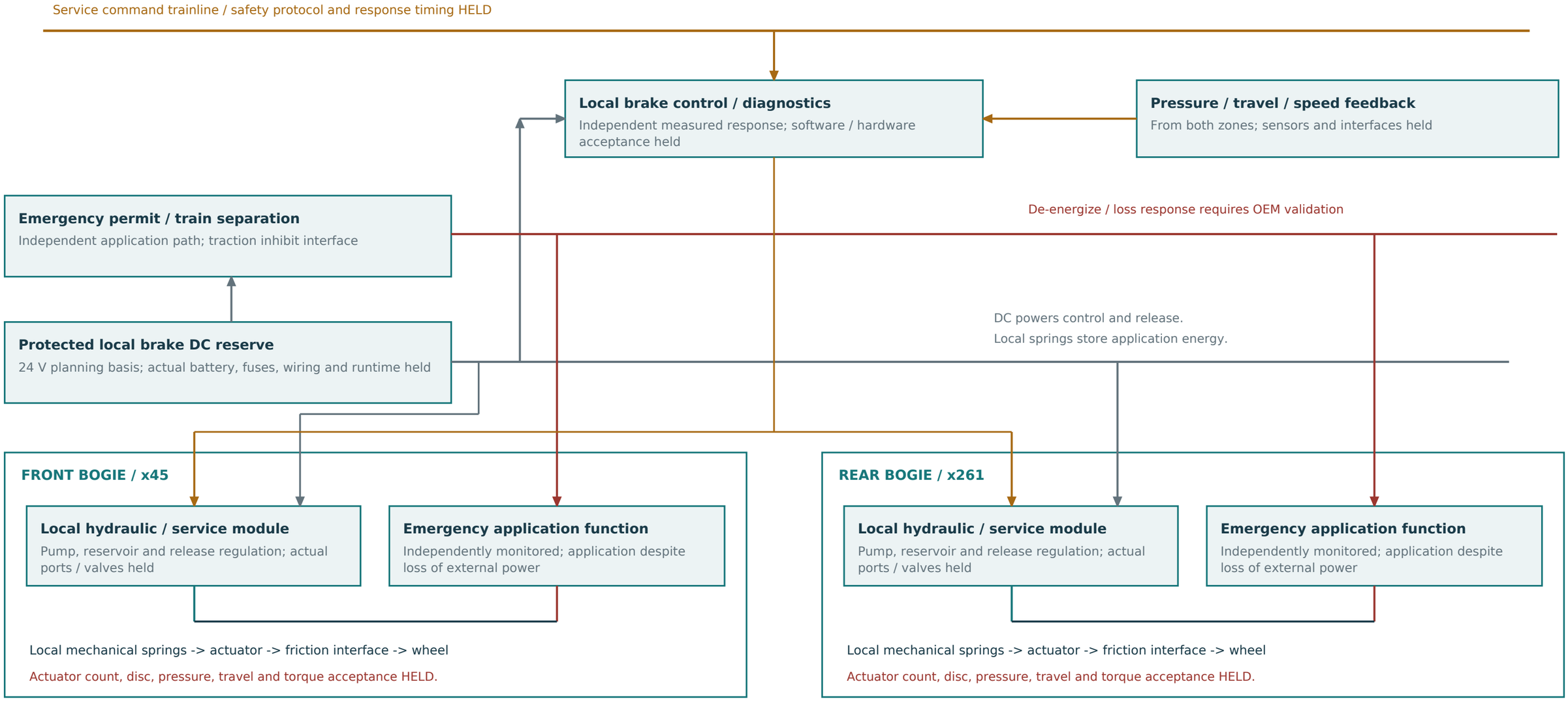
17 x 11 in / R0

PULSIPHER FARMS / CHRISTMAS RAILWAY

Hydraulic brake architecture: local application energy and monitored zones

Functional inquiry family: spring-applied / hydraulically released service modulation. OEM duty, brake balance and fault evidence must support it; actual valve circuits and port details remain HELD.

S1 / COMMAND, PERMIT AND ENERGY PATHS - NOT A VALVE CIRCUIT



Shared actuator / disc are common-cause items, not a second independent brake system. No intercar hydraulic service hoses. Auxiliary recovery release requires separate OEM isolation / restoration design so trapped pressure cannot defeat application.

VERIFIED / source data CALCULATED / stated inputs STUDY / proposed geometry HELD / required before release

PRELIMINARY ENGINEERING DEFINITION - NOT FOR FABRICATION OR CONSTRUCTION

PF-TRAIN-S3F-R0 | 2026-10-18 | References: systems_f / functional hydraulic architecture

S01

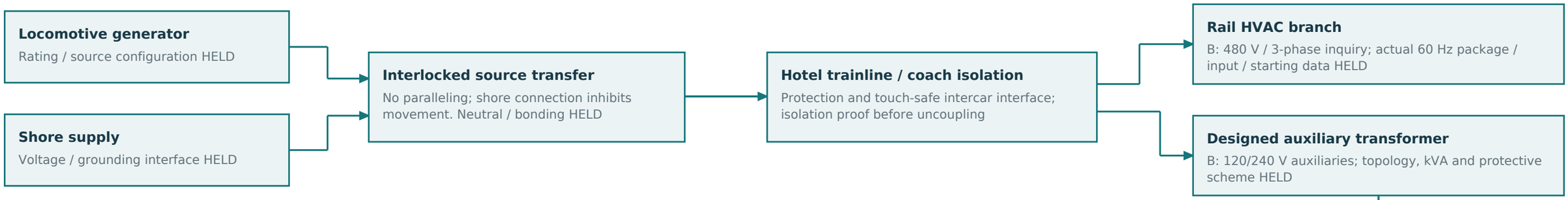
17 x 11 in / R0

PULSIPHER FARMS / CHRISTMAS RAILWAY

Electrical one-line: hotel power, local brake reserve and emergency lighting

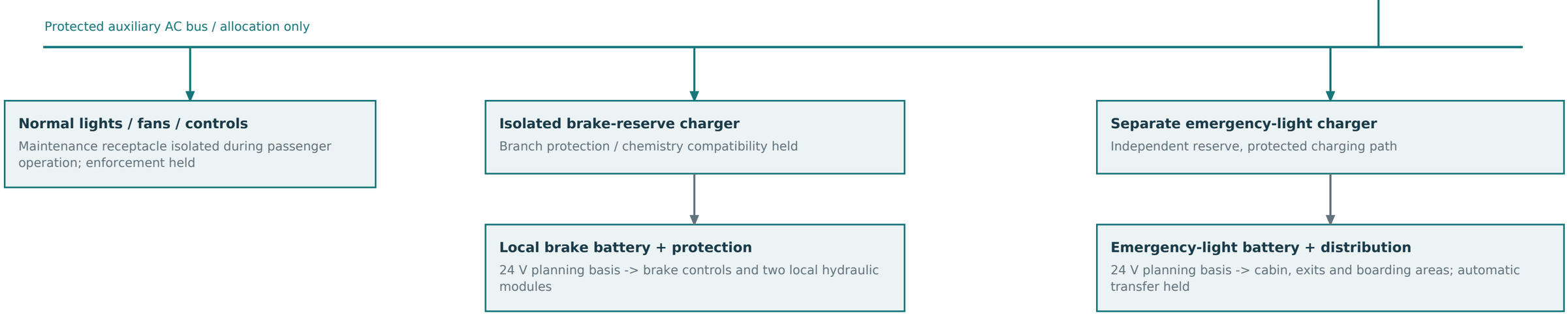
Architecture B is an inquiry preference, not a selected voltage system. Actual loads, fault current, protective coordination, bonding, connectors, cable sizes and transfer details require electrical design.

E1 / HOTEL AC - TWO SOURCES, CONTROLLED TRANSFER, PROTECTED DISTRIBUTION



Alternative A: 120/240 V split-phase hotel system with compatible equipment. The 400 V catalog rail HVAC reference is not a direct 480 V load. Resolve voltage / frequency and starting demand before source or connector selection.

E2 / AUXILIARIES AND SEPARATE LOCAL DC RESERVES



Protective bonding continuity must cross flexible intercar interfaces; do not use rails or coupler contact as the sole protective conductor. Neutral / PE design HELD.

VERIFIED / source data | CALCULATED / stated inputs | STUDY / proposed geometry | HELD / required before release

PRELIMINARY ENGINEERING DEFINITION - NOT FOR FABRICATION OR CONSTRUCTION

PF-TRAIN-S3F-R0 | 2026-10-18 | References: systems_f / functional power architecture

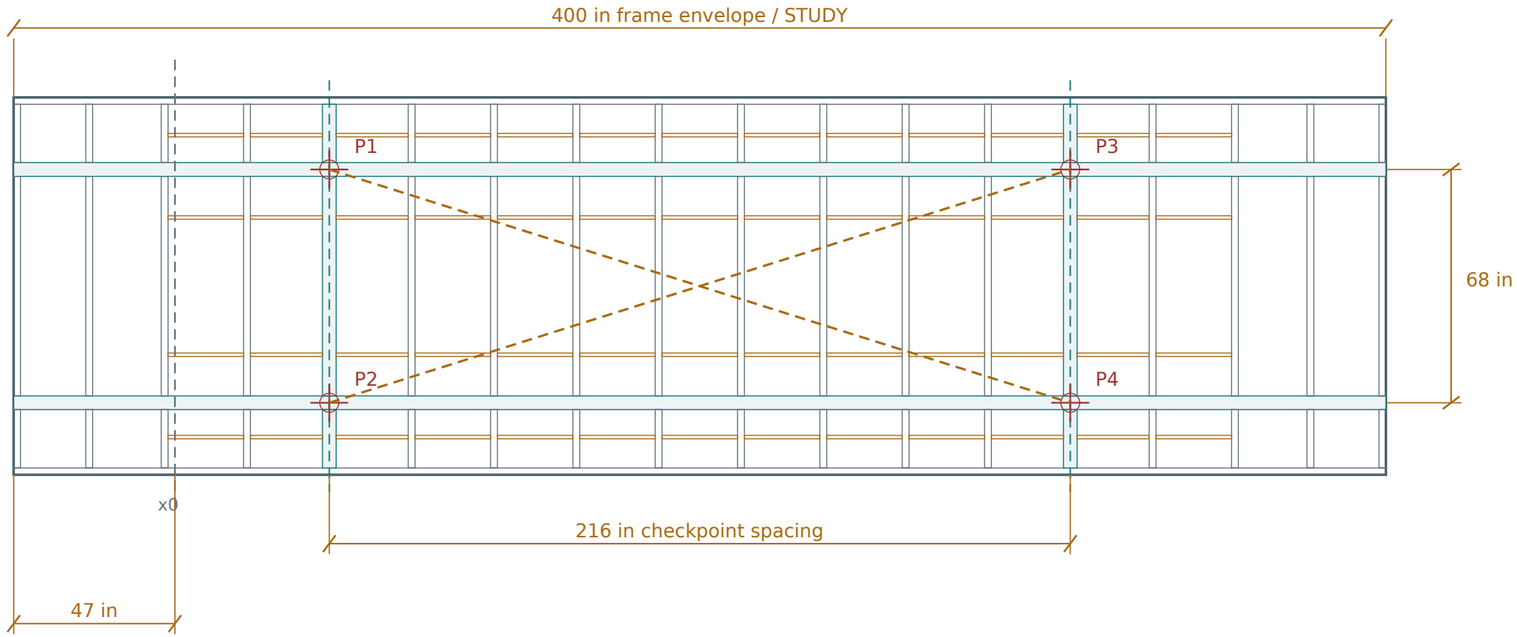
E01

17 x 11 in / R0

Assembly datum and controlled fabrication sequence

F1 retains E1 frame geometry. Dimensioned measurement references are not fixture supports, approved weld sequences, lifting points or machining instructions.

F1 / FRAME DATUM TRANSFER AND SURVEY CHECKPOINTS



Target P1-P4 z = 31.003937 in. Equal nominal diagonals = 226.4509 in.

Measure unloaded and after each released joining / turning step; allowable deviation HELD.

F2 / PHYSICAL DATUM CONTRACT

Datum	Study reference
A	Steel bearing plane z31.003937
B	y0 from main centrelines y +/-34
C	Front frame face x-47
x0	47 in aft of datum C

P1-P4 mark existing girder / bolster intersections for measurement. Do not place jacks or clamps at these symbols without a reviewed support design.

Fixture contact positions, restraint forces, weld details, distortion allowance, NDE methods and acceptance limits remain HELD. All members remain no-drill / no-mount until the specific detail is released.

F3 / BUILD TRAVELER - RECORD EACH LAYER BEFORE IT IS COVERED

01 MATERIAL / CUT ID
Match released sections and certificates; mark every member and preserve traceability.

02 DRY FIT / DATUM
Verify part identity, intersections, diagonals and service access before joining.

03 JOIN / INSPECT
Use released WPS / joint map and sequence; record dimensional and specified NDE results.

04 CLOSE / FIT OUT
Accept coatings, bonds and hidden interfaces before floor/body closure; recheck access.

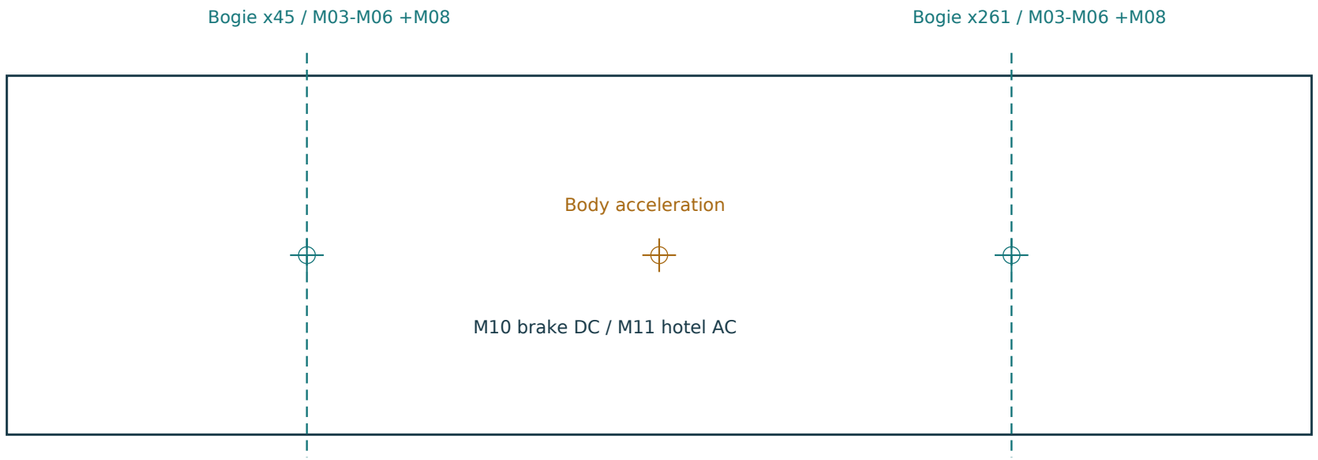
Farm work: identification, measured records and released nonstructural assembly. Specialist release governs load-bearing joins, lifting, electrical and brake work.

VERIFIED / source data CALCULATED / stated inputs STUDY / proposed geometry HELD / required before release

First-article commissioning: installed measurements and evidence packages

All physical tests remain unperformed. Actual protocols, test loads, acceptance limits, restraint, instruments and witness requirements must be accepted before execution.

T1 / MEASUREMENT ZONES - SENSOR MOUNTS REMAIN HELD



M01 common time; M02 independent speed/distance; M12 air / temperature / humidity;

M13 emergency lighting; M14 +M17 floor / support motion; M15 seam movement.

T2 / SPECIALIST WORK PACKAGES

ID	Package	Lead
SF-W01	Installation / access	Vehicle + maintenance
SF-W02	Hydraulic / controls	Brake authority + loco
SF-W03	Source / circuits	Electrical engineer
SF-W04	Thermal / airflow	HVAC + body engineer
SF-W05	Installed functions	Test director / integrators

Before structural release: obtain Building Official classification and applicable standard editions. If Chapter 22.16 applies, coordinate the Nevada principal design professional and the approved fabrication / inspection / commissioning route.

T3 / ACCEPTED RECORDS MUST MATCH THE AS-BUILT CONFIGURATION

Record family	Installed evidence	Required review basis
E01 /E02	Mass, wheel loads, floor / hardpoints	Calibrated mass and independent reference; released test load / limits
E03-E09	Brake functions and vehicle motion	Measured force / pressure / travel, empty/full balance and fault evidence
E10-E12	Power, HVAC and blackout	Transfer/protection, delivered thermal / airflow, lighting endurance
E14 then E13	Installed boarding then participant exercises	Accepted bridge/lift installation before supervised evacuation practice

A first-coach acceptance does not release the full consist. If county Chapter 22.16 applies, provide the approved Qualified Individual, county commissioning and Certificate of Operation pathway.

VERIFIED / source data CALCULATED / stated inputs STUDY / proposed geometry HELD / required before release